

FOUNDATION FOR VIRTUAL AND AUGMENTED REALITY SYSTEMS

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PRE-REQUISITES :

Science/engineering background preferable. Knowledge in software system development, data structure, algorithms and computer graphics will be helpful although not mandatory.

INTENDED AUDIENCE :

UG/PG/PhD students in Computer Science & Engineering and/or allied disciplines. Also useful for software/system developers, with interest in extended reality.

INDUSTRY SUPPORT :

Application-focused companies should be interested, cutting across domains. Some of the relevant industries include,

1. Microsoft
2. Alphabet
3. Meta
4. Samsung
5. Many startups
6. Companies focused on hardware (GPU, headsets) – e.g. Nvidia, Qualcomm etc

COURSE OUTLINE :

Extended Reality (XR), covering the areas of Virtual, Augmented and Mixed Reality technologies, has become quite popular nowadays. The reason is primarily the advancements in relevant hardware, which made such technologies affordable. As a result, there is huge interest in building applications of such technologies in various domains, the most notable example being the idea of Metaverse. This course is focused on explaining the foundational aspects of the technology in a lucid manner, through a practical case study. The topics include the basic concepts of XR, the hardware and software used, the issues (both psycho-physiological and technological) and how to address those from the user's and developer's perspective, stages of system development and finally, optimization issues and approaches to resolve. Each concept is explained in details, with the help of the use case, whenever appropriate. The overall aim of the course is to impart foundational knowledge to the learner to understand and implement an XR system with confidence.

ABOUT INSTRUCTOR :

Prof. Samit Bhattacharya is an Associate Professor in the Dept of Computer Science and Engineering, IIT Guwahati, with more than 16 years of teaching and research experience. He has taught a plethora of courses at various levels both at the IIT Guwahati and outside, including courses related to this one. He is also an active researcher in virtual, augmented & mixed reality systems & technologies. So far, he has graduated 7 PhDs (including one in this area) and guided 20+ MTech and 40+ BTech projects (several of which are related to the topics in this course). He has 85 publications under his credit including books, book chapters, patent, peer-reviewed journals and conference proceedings (many of those are related to the course topics). He has also managed research & consultancy grants worth over 10.5 crores INR as PI & Co-PI, obtained from both Govt and industry sources in India & abroad (including 2 related to training & technology development in the areas related to this course). He actively reaches out to academia and industry at the national and international level through academic visits, collaborations, invited lectures, NPTEL MOOCs, online certification programs, FDPs and EDPs. More about him can be found at https://www.iitg.ac.in/samit/

COURSE PLAN :

Week 1: Introduction
1. General introduction
2. Application domains & system categories
3. Definitions

Week 2: Introduction
4. Historical evolution
5. Issues & challenges
6. Cybersickness

Week 3: XR Hardware & Software
7. VR Hardware
8. AR Hardware
9. Understanding Tracking I

Week 4: XR Hardware & Software
10. Understanding Tracking II
11. VR Software
12. AR Software

Week 5: Human factors in XR experience
13. XR as Interactive Systems
14. Human Factors in XR
15. Visual Perception & Displays I

Week 6: Human factors in XR experience
16. Visual Perception & Displays II
17. Auditory Sense & Immersive Experience
18. Other Senses & Full Immersion

Week 7: Building of an XR System –understanding with a case study
19. Interactive System Development Life Cycle Stages
20. Introduction to the Case Study
21. Requirement Gathering I
22. Requirement Gathering II

Week 8: Building of an XR System –understanding with a case study
23. System Design I
24. System Design II
25. System Design III

Week 9: Building of an XR System –understanding with a case study
26. System Implementation I
27. System Implementation II
28. System Implementation III

Week 10: System Optimization
29. Optimizing User Experience I – Basic Idea
30. Optimizing User Experience II – Navigation Time
31. Optimizing User Experience III – Navigation Path

Week 11: System Optimization
32. Optimizing Resources I – Basic Idea
33. Understanding Resource Requirements – Introduction to Computer Graphics I
34. Understanding Resource Requirements – Introduction to Computer Graphics II

Week 12: Conclusion
35. Understanding Resource Requirements – Introduction to Computer Graphics III
36. Resource Optimization Techniques
37. Concluding Remarks